Arista Access Node MAR-ACC-04 QoS Congestion Event Resolution SOP

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3675

Description

This Standard Operating Procedure outlines the steps to diagnose and resolve Quality of Service (QoS) congestion events on the Arista Access Node MAR-ACC-04, located at the Marathahalli Core site. The failure domain is characterized by microbursts, interface tail drops, and QoS queue starvation, manifesting as a Major alarm (ALARM_3675). The root cause of this issue lies in the interaction between the Access Node's QoS policies and the underlying hardware architecture. Specifically, the Arista 7150 series switch uses a shared buffer architecture, where packets from multiple queues are stored in a common buffer. When a particular queue experiences a surge in traffic, it can lead to buffer overflows, causing packets to be dropped, and resulting in QoS queue starvation. Furthermore, the switch's congestion avoidance mechanism may not be effective in mitigating microbursts, exacerbating the issue. To ensure proper QoS configuration and prevent congestion events, it is crucial to monitor and adjust the QoS policies, particularly the queue settings and buffer sizes, to match the expected traffic patterns and avoid over-subscription.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to access the MAR-ACC-04 node.
- Verify the status of the redundant link and confirm it is operational.
- Check the power supplies and cooling systems to ensure they are functioning correctly.

Potential Root Causes

Cause	Probability
Inadequate QoS policy configuration	High
Insufficient buffer sizes	Medium
Microburst events caused by unexpected traffic patterns	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos queue statistics, Expected Output: Queue statistics, including packet drops and buffer overflows, Description: This command displays the queue statistics, helping to identify which queues are experiencing congestion and packet drops.

Command: show buffer statistics, Expected Output: Buffer statistics, including buffer overflows and packet drops, Description: This command displays the buffer statistics, providing insight into the buffer overflows and packet drops caused by queue congestion.
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```
Command: show interface counters, Expected Output: Interface counters, including packet drops and errors, Description: This command displays the interface counters, helping to identify which interfaces are experiencing packet drops and errors due to congestion.
```

Resolution Steps

Step 1: Step: Adjust QoS policy configuration, Command: configure terminal qos policy-map policy-name class class-name queue queue-name, Description: Adjust the QoS policy configuration to prioritize traffic and prevent over-subscription. This may involve adjusting the queue settings, buffer sizes, or queue scheduling algorithms.

Step 2: Step: Increase buffer sizes, Command: configure terminal buffer size size, Description: Increase the buffer sizes to prevent overflows and packet drops. The new buffer size should be sufficient to handle the expected traffic patterns.

Step 3: Step: Implement congestion avoidance mechanisms, Command: configure terminal congestion avoidance mechanism, Description: Implement congestion avoidance mechanisms, such as traffic shaping or policing, to prevent microburst events and maintain QoS queue starvation.

Post-Check Verification

```
Command: show qos queue statistics, Expected Output: Queue statistics showing reduced packet drops and buffer overflows, Description: Verify that the QoS policy adjustments and buffer size increases have resolved the congestion event by checking the queue statistics.

Command: show buffer statistics, Expected Output: Buffer statistics showing reduced buffer overflows and packet drops, Description: Verify that the congestion avoidance mechanisms have been effective in preventing microburst events and maintaining QoS queue starvation by checking the buffer statistics.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure terminal default qos policy-map policy-name class class-name queue queue-name, Description: Restore the original QoS policy configuration to prevent unintended behavior.

- Step: Restore original buffer sizes, Command: configure terminal buffer size original-size, Description: Restore the original buffer sizes to prevent over-subscription and congestion events.

- Step: Disable congestion avoidance mechanisms, Command: configure terminal no congestion avoidance mechanism, Description: Disable the congestion avoidance mechanisms to prevent unintended behavior and maintain normal operation.